

CO5 – AT1

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Q1. Uncertainty of Point Estimates

For a manufacturing firm's Plant Manager, build an error-bar/interval chart of the point estimate for each entity for average product lifespan (months) across 5 lines. Then analyze which entities' intervals overlap versus which are clearly separated, and state what this implies about whether their true values actually differ, and include one written recommendation for the Plant Manager.

Code:

```
import numpy as np
import matplotlib.pyplot as plt
import matplotlib.font_manager as fm

fm.fontManager.addfont('/usr/share/fonts/truetype/liberation/LiberationSerif-
Regular.ttf')
plt.rcParams['font.family'] = 'Liberation Serif'
np.random.seed(1)
lines = ["Line A", "Line B", "Line C", "Line D", "Line E"]

# simulated sample data: each line has a small sample of measured product
# lifespans (months) from quality-control testing
n_per_line = 12
true_means = [58, 61, 60.5, 47, 46]
true_sd = [4, 4.5, 4.2, 5, 5.5]
samples = [np.random.normal(m, s, n_per_line) for m, s in zip(true_means, true_sd)]

point_estimates = [s.mean() for s in samples]
sds = [s.std(ddof=1) for s in samples]
sems = [sd / np.sqrt(n_per_line) for sd in sds]

# 95% CI using t-distribution approx (t ~ 2.20 for df=11)
t_val = 2.20
ci95 = [t_val * sem for sem in sems]

fig, ax = plt.subplots(figsize=(7, 4.5))
x = np.arange(len(lines))

ax.errorbar(x, point_estimates, yerr=ci95, fmt='o', color='black',
            ecolor='black', elinewidth=1.5, capsize=6, markersize=7)

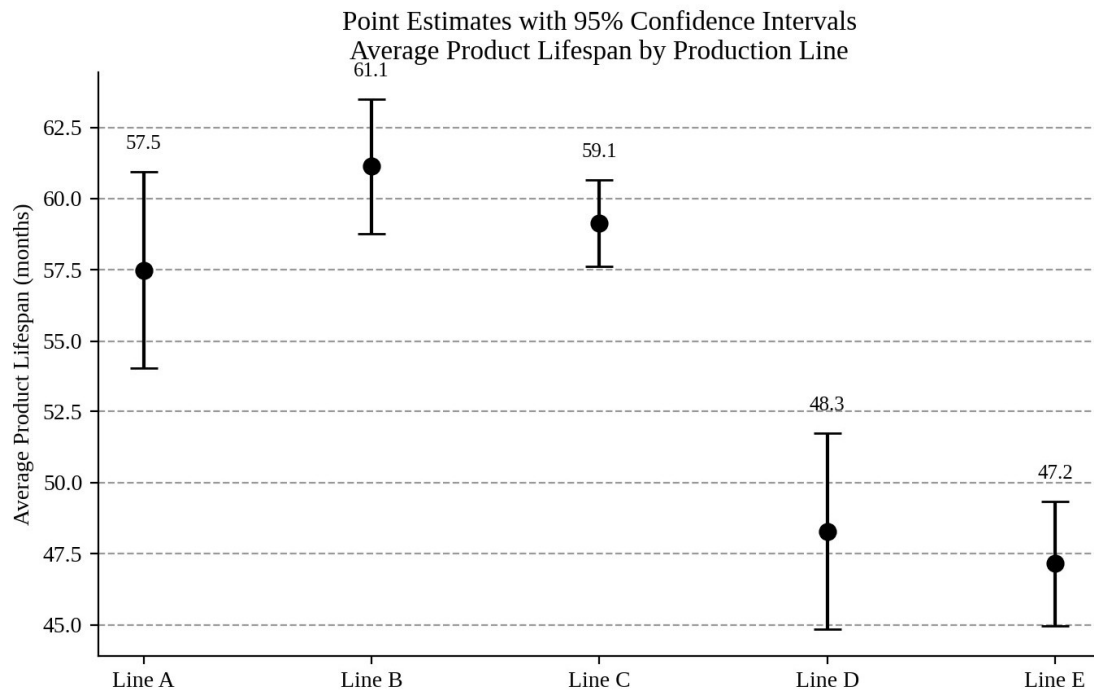
ax.set_xticks(x)
ax.set_xticklabels(lines)
ax.set_ylabel("Average Product Lifespan (months)")
ax.set_title("Point Estimates with 95% Confidence Intervals\nAverage Product Lifespan by Production Line")
ax.grid(axis='y', linestyle='--', alpha=0.4, color='black')
for spine in ['top', 'right']:
    ax.spines[spine].set_visible(False)

for xi, yi, ci in zip(x, point_estimates, ci95):
    ax.text(xi, yi + ci + 0.8, f"{yi:.1f}", ha='center', fontsize=9)

plt.tight_layout()
plt.savefig("q1_lifespan_intervals.png", dpi=200)
plt.close()

# print numbers for reference / to embed in analysis
for name, pe, ci in zip(lines, point_estimates, ci95):
    print(f"{name}: mean={pe:.2f}, 95% CI = [{pe-ci:.2f}, {pe+ci:.2f}]")
```

Output:



Q2. Uncertainty in Trend Estimation

For an agri-cooperative's Coop President, build a trend line over time with a shaded confidence band around it for estimated crop yield (tonnes/hectare) across 7 districts. Then analyze where the band is narrowest and widest, and explain why (e.g., data density, extrapolation) using the specific portions of the trend, and include one written recommendation for the Coop President.

Code:

```
import numpy as np
import matplotlib.pyplot as plt
import matplotlib.font_manager as fm

fm.fontManager.addfont('/usr/share/fonts/truetype/liberation/LiberationSerif-Regular.ttf')
plt.rcParams['font.family'] = 'Liberation Serif'
np.random.seed(2)

years_hist = np.arange(2015, 2025) # actual recorded years (10 years)
years_future = np.array([2025, 2026]) # extrapolated years
years_all = np.concatenate([years_hist, years_future])

n_districts = 7
# simulate yearly yield readings from 7 districts (more districts reporting in recent years than in early years -> denser data recently)
reports_per_year = {y: n for y, n in zip(years_hist, [3, 4, 4, 5, 6, 6, 7, 7, 7])}
true_trend = 2.0 + 0.12 * (years_hist - 2015) # underlying upward trend

means, ci_low, ci_high = [], [], []
for i, y in enumerate(years_hist):
    n = reports_per_year[y]
    draws = np.random.normal(true_trend[i], 0.35, n)
    m = draws.mean()
    sem = draws.std(ddof=1) / np.sqrt(n)
    if n > 1 else 0.35
    means.append(m)
```

```

ci_low.append(m - 1.96 * sem)    ci_high.append(m +
1.96 * sem)

# extrapolated years: no real data -> uncertainty widens sharply
last_trend = true_trend[-1] for j, y in enumerate(years_future):    m =
last_trend + 0.12 * (j + 1)    sem = 0.35 * (2.2 + j * 1.3) # growing
uncertainty with extrapolation    means.append(m)
ci_low.append(m - 1.96 * sem)    ci_high.append(m + 1.96 * sem)

means = np.array(means) ci_low = np.array(ci_low) ci_high =
np.array(ci_high) fig, ax = plt.subplots(figsize=(7.5, 4.5)) ax.plot(years_all,
means, color='black', linewidth=1.8, marker='o', markersize=4,
label="Estimated mean yield") ax.fill_between(years_all, ci_low, ci_high,
color='black', alpha=0.15,          label="95% confidence band")

# mark boundary between observed and extrapolated region ax.axvline(2024.5,
color='black', linestyle=':', linewidth=1) ax.text(2024.6, ax.get_ylim()[1]*0 +
means.max() + 0.4, "Extrapolated\n(no district data)",          fontsize=8, ha='left')

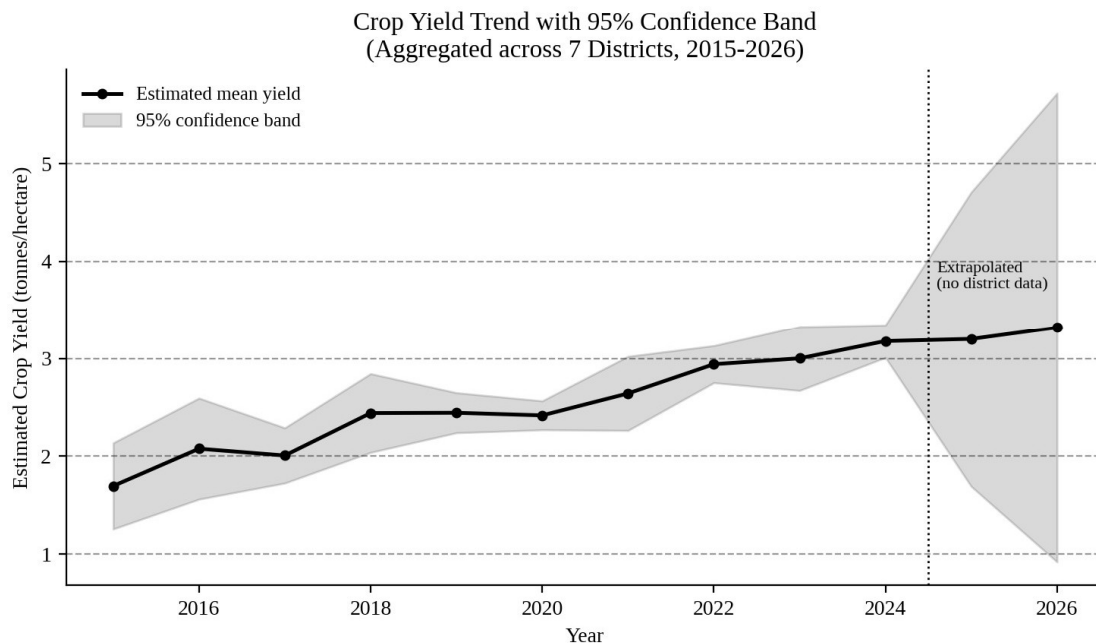
ax.set_xlabel("Year") ax.set_ylabel("Estimated Crop Yield (tonnes/hectare)") ax.set_title("Crop Yield
Trend with 95% Confidence Band\n(Aggregated across 7 Districts, 2015-2026)") ax.grid(axis='y',
linestyle='--', alpha=0.4, color='black') for spine in ['top', 'right']:
    ax.spines[spine].set_visible(False)
ax.legend(frameon=False, loc='upper left', fontsize=9)

plt.tight_layout()
plt.savefig("q2_yield_trend_band.png", dpi=200)
plt.close()

band_width = ci_high - ci_low for y, bw, n in zip(years_all,
band_width,          list(reports_per_year.values())) +
['extrap','extrap']):    print(f"{y}: band width={bw:.2f},
districts reporting={n}")

```

Output:



Q3. Hypothetical Outcome Plots

For an agri-cooperative's Coop President, build a Hypothetical Outcome Plot (HOP) - or a static fan-chart approximation using many simulated draws - of the uncertain estimate for estimated crop yield (tonnes/hectare) across 7 districts. Then analyze how the spread of individual draws communicates uncertainty to a non-technical viewer more intuitively than a single static error bar, citing what a viewer would perceive from a few draws, and include one written recommendation for the Coop President.

Code:

```
import numpy as np
import matplotlib.pyplot as plt
import matplotlib.font_manager as fm

fm.FontManager.addfont('/usr/share/fonts/truetype/liberation/LiberationSerif-Regular.ttf')
plt.rcParams['font.family'] = 'Liberation Serif'
np.random.seed(3)

districts = ["D1", "D2", "D3", "D4", "D5", "D6", "D7"]
district_names = ["Anpur", "Belgola", "Chikodi", "Devihalli", "Erode Rural", "Falna", "Gulbura"]

means = [3.2, 3.6, 2.9, 4.1, 3.4, 3.0, 3.8]
sds = [0.55, 0.30, 0.70, 0.40, 0.65, 0.50, 0.35]
n_draws = 300

fig, ax = plt.subplots(figsize=(7.5, 4.8))

x = np.arange(len(districts))
rng = np.random.default_rng(3)

for i, (m, s) in enumerate(zip(means, sds)):
    draws = rng.normal(m, s, n_draws)
    jitter = rng.uniform(-0.28, 0.28, n_draws)
    ax.scatter(x[i] + jitter, draws, color='black', alpha=0.08, s=10, linewidths=0)
    # mark the mean of the draws
    ax.plot([x[i]-0.32, x[i]+0.32], [draws.mean(), draws.mean()], color='black', linewidth=2)

# highlight a handful of individual "single draw" outcomes across districts
sample_draw_idx = 7
single_draw = [rng.normal(m, s) for m, s in zip(means, sds)]
ax.scatter(x, single_draw, facecolors='none', edgecolors='black', s=90, linewidths=1.6, zorder=5, label="One single hypothetical outcome")

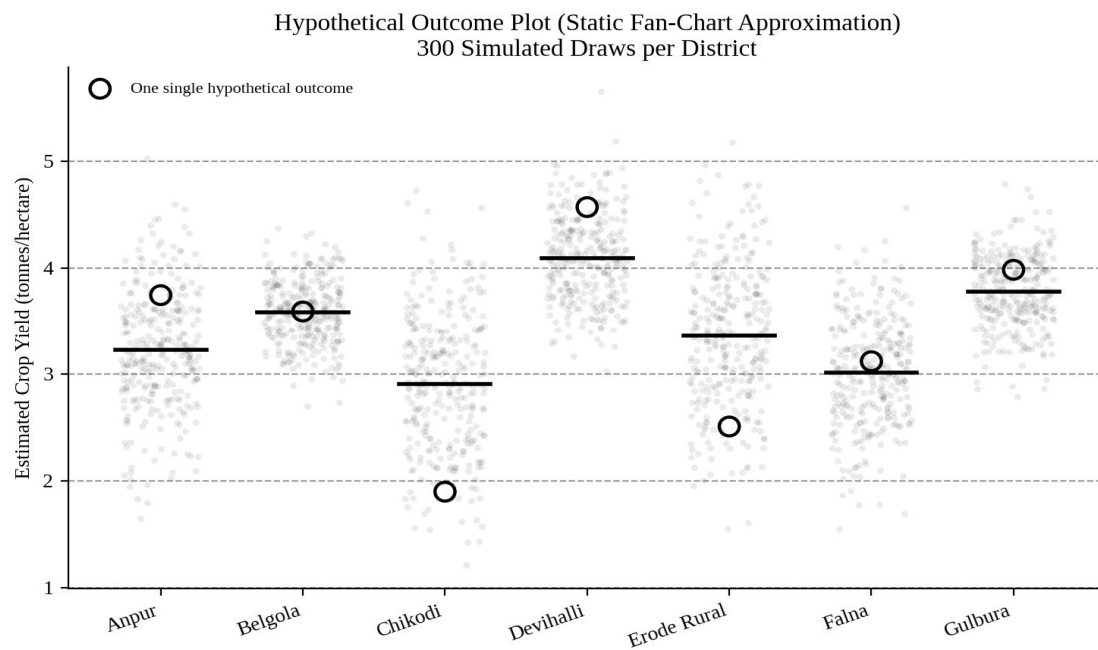
ax.set_xticks(x)
ax.set_xticklabels(district_names, rotation=20, ha='right')
ax.set_ylabel("Estimated Crop Yield (tonnes/hectare)")
ax.set_title("Hypothetical Outcome Plot (Static Fan-Chart Approximation)\n" f"{n_draws} Simulated Draws per District")

ax.grid(axis='y', linestyle='--', alpha=0.4, color='black')
for spine in ['top', 'right']:
    ax.spines[spine].set_visible(False)
ax.legend(frameon=False, loc='upper left', fontsize=8)

plt.tight_layout()
plt.savefig("q3_yield_hop_fanchart.png", dpi=200)
plt.close()
```

```
for name, m, s in zip(district_names, means, sds):  
    print(f'{name}: mean={m}, sd={s}')
```

Output:



Q4. Bootstrapping and Confidence Intervals

For an agri-cooperative's Coop President, build a bootstrap resampling procedure (at least 1000 resamples) on a small sample, with a histogram of the resulting bootstrap estimates and a 95% confidence interval marked for estimated crop yield (tonnes/hectare) across 7 districts. Then analyze how wide the resulting CI is relative to the point estimate, and what that says about how much the original sample size limits confidence, and include one written recommendation for the Coop President.

Code:

```
import numpy as np
import matplotlib.pyplot as plt
import matplotlib.font_manager as fm

fm.FontManager.addfont('/usr/share/fonts/truetype/liberation/LiberationSerif-
Regular.ttf')
plt.rcParams['font.family'] = 'Liberation Serif'
rng = np.random.default_rng(4)

district_names = ["Anpur", "Belgola", "Chikodi", "Devihalli", "Erode Rural",
                  "Falna", "Gulbura"]

# small original sample: one yield reading (tonnes/hectare) per district
sample = np.array([3.1, 3.6, 2.8, 4.0, 3.3, 2.9, 3.7])
n = len(sample)

n_boot = 5000
boot_means = np.empty(n_boot)
for i in range(n_boot):
    resample = rng.choice(sample, size=n, replace=True)
    boot_means[i] = resample.mean()

point_estimate = sample.mean()
ci_low, ci_high = np.percentile(boot_means, [2.5, 97.5])

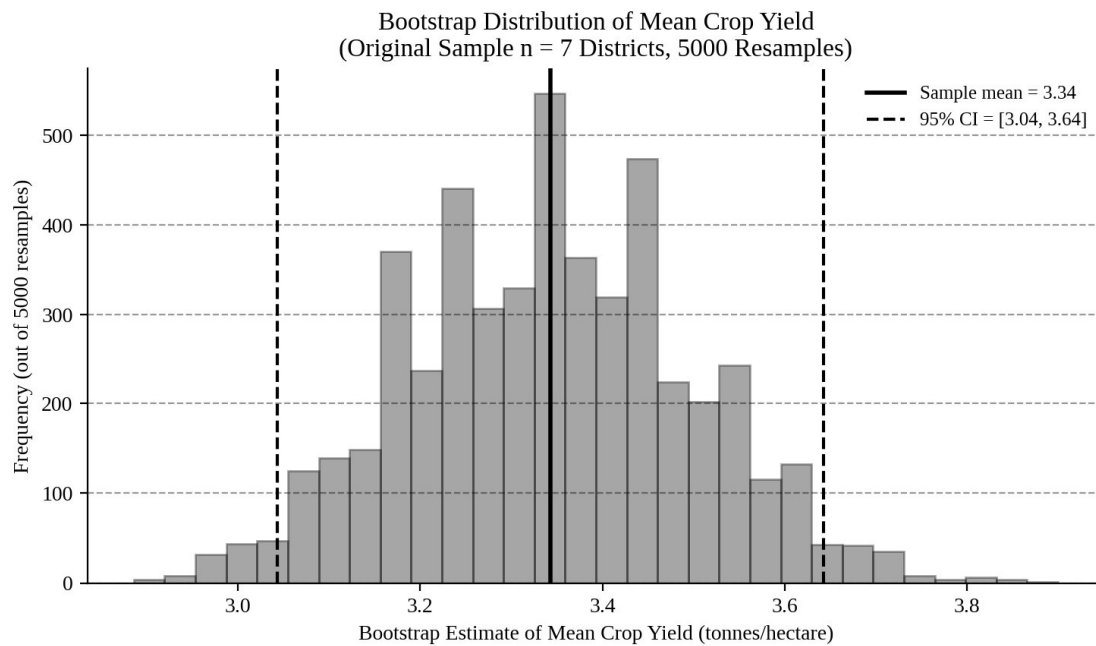
fig, ax = plt.subplots(figsize=(7.5, 4.5))
ax.hist(boot_means, bins=30, color='black', edgecolor='black', alpha=0.35)
ax.axvline(point_estimate, color='black', linewidth=2, linestyle='-',
            label=f"Sample mean = {point_estimate:.2f}")
ax.axvline(ci_low, color='black', linewidth=1.5, linestyle='-',
            label=f"95% CI = [{ci_low:.2f}, {ci_high:.2f}]")
ax.axvline(ci_high, color='black', linewidth=1.5, linestyle='--')

ax.set_xlabel("Bootstrap Estimate of Mean Crop Yield (tonnes/hectare)")
ax.set_ylabel("Frequency (out of 5000 resamples)")
ax.set_title(f"Bootstrap Distribution of Mean Crop Yield\n(Original Sample n = {n} Districts, 5000 Resamples)")
ax.grid(axis='y', linestyle='--', alpha=0.4, color='black')
for spine in ['top', 'right']:
    ax.spines[spine].set_visible(False)
ax.legend(frameon=False, fontsize=9)

plt.tight_layout()
plt.savefig("q4_bootstrap_ci.png", dpi=200)
plt.close()

print(f"Original sample (n={n}): {sample}")
print(f"Point estimate (mean): {point_estimate:.3f}")
print(f"95% Bootstrap CI: [{ci_low:.3f}, {ci_high:.3f}]")
print(f"CI width: {ci_high - ci_low:.3f}")
print(f"CI width as % of point estimate: {(ci_high-ci_low)/point_estimate*100:.1f}%")
```

Output:



Q5. Principles of Effective Visualization

For an agri-cooperative's Coop President, build a chart built by a colleague without reference to any design principles (arbitrary colours, no clear title, inconsistent scales) for estimated crop yield (tonnes/hectare) across 7 districts. Then analyze the chart against the core principles of effective visualization (clarity, accuracy, honesty/no distortion, simplicity, accessibility) and identify which principles are violated, and include one written recommendation for the Coop President.

Code:

```
import numpy as np
import matplotlib.pyplot as plt
import matplotlib.font_manager as fm

fm.fontManager.addfont('/usr/share/fonts/truetype/liberation/LiberationSerif-Regular.ttf')
plt.rcParams['font.family'] = 'Liberation Serif'

district_names = ["Anpur", "Belgola", "Chikodi", "Devihalli", "Erode Rural",
                  "Falna", "Gulbura"]
yield_vals = [3.1, 3.6, 2.8, 4.0, 3.3, 2.9, 3.7]

# arbitrary, clashing colours with no meaning
colors = ['#ff0000', '#00ff00', '#0000ff', '#ffcc00', '#ff00ff', '#00ffff', '#8b0000']

fig, ax = plt.subplots(figsize=(7.5, 4.5))
bars = ax.bar(district_names, yield_vals, color=colors, edgecolor='black', linewidth=2)

# no clear/informative title
# inconsistent, truncated y-axis that exaggerates differences
ax.set_ylim(2.5, 4.2)

# no axis label on y-axis, unclear units
ax.set_xticklabels(district_names, rotation=45, ha='right', fontsize=11)
```



```

# unnecessary 3D-looking shadow / chart-junk via duplicated offset
bars for i, v in enumerate(yield_vals): ax.bar(i + 0.08, v, width=0.7,
color='grey', alpha=0.3, zorder=0)

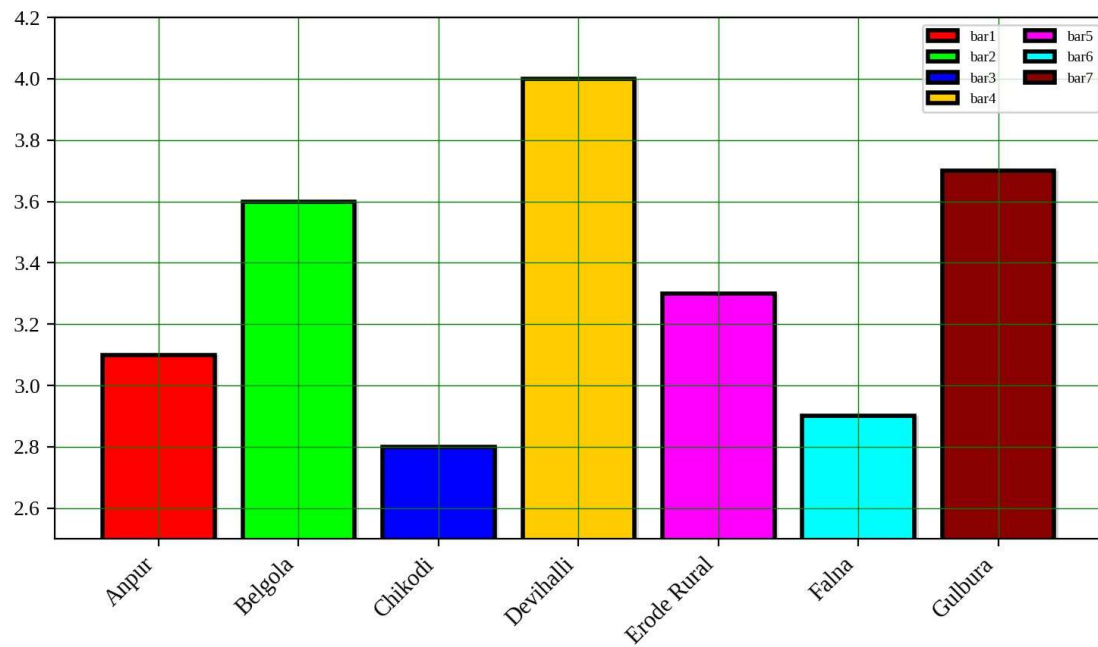
# random gridlines on both axes, cluttered ax.grid(True,
which='both', color='green', linestyle='-', linewidth=0.5)

# legend that adds no information ax.legend(bars, [f"bar{i+1}" for i in range(7)],
loc='upper right', fontsize=7, ncol=2)

plt.tight_layout()
plt.savefig("q5_bad_chart.png", dpi=200)
plt.close()
print("bad chart generated")

```

Output:



Q1 Uncertainty of Point Estimates

Lines A, B and C overlap heavily, so we can't say their true lifespans differ. Lines D and E overlap each other but sit clearly below and separate from A/B/C, showing a real difference. Recommendation: Plant Manager should investigate Lines D and E's process/equipment, not chase differences within A/B/C.

Q2 Uncertainty in Trend Estimation

The confidence band is narrowest around 2019-2020 and 2024, when most of the 7 districts reported data. It is widest in 2025-2026, since those years are extrapolated with no real data.

Recommendation: Coop President should trust 2019-2024 figures for planning but avoid using the 2025-2026 projection without fresh data.

Q3 Hypothetical Outcome Plots

Districts like Chikodi and Erode Rural show widely scattered draws, while Belgola and Gulbura show tighter, more predictable clouds.

A single highlighted draw landing far from the average makes uncertainty feel real, unlike a static error bar. Recommendation: Coop President should use HOP/fan-chart visuals in farmer meetings for more intuitive risk communication.

Q4 Bootstrapping and Confidence Intervals

The bootstrap 95% CI (3.04-3.64 tonnes/hectare) is about 18% wide relative to the point estimate (3.34), which is fairly wide. This reflects the small original sample of just 7 readings, limiting how precisely the true yield can be pinned down.

Recommendation: Coop President should collect more yield readings per district to narrow future confidence intervals.

Q5 Principles of Effective Visualization

The chart violates clarity (no title/axis label), honesty (truncated y-axis exaggerates differences), simplicity (shadow bars as chart-junk), and accessibility (clashing arbitrary colors, cluttered gridlines). Recommendation: Coop President should require charts to use

a zero-based axis, one neutral color, and a clear labeled title before sharing with members.